

Bayesian inference for a wave-front model of the neolithization of Europe

Andrew W. Baggageley,^{*} Graeme R. Sarson, Anvar Shukurov, Richard J. Boys, and Andrew Golightly
School of Mathematics and Statistics, Newcastle University, Newcastle upon Tyne, NE1 7RU, England, United Kingdom
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We consider a wave-front model for the spread of neolithic culture across Europe, and use Bayesian inference techniques to provide estimates for the parameters within this model, as constrained by radiocarbon data from southern and western Europe. Our wave-front model allows for both an isotropic background spread (incorporating the effects of local geography) and a localized anisotropic spread associated with major waterways. We introduce an innovative numerical scheme to track the wave front, and use Gaussian process emulators to further increase the efficiency of our model, thereby making Markov chain Monte Carlo methods practical. We allow for uncertainty in the fit of our model, and discuss the inferred distribution of the parameter specifying this uncertainty, along with the distributions of the parameters of our wave-front model. We subsequently use predictive distributions, taking account of parameter uncertainty, to identify radiocarbon sites which do not agree well with our model. These sites may warrant further archaeological study or motivate refinements to the model.

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I. INTRODUCTION

The transition from hunter-gathering to early farming—signifying the start of the neolithic era in traditional archaeological terminology—was one of the most important steps made by humanity in developing the complex modern societies that exist today. The mechanism of the spread throughout Europe of neolithic farming techniques, which developed in the Near East around 12 000 years ago, remains an important and fascinating question. The relative importance of “cultural” versus “demic” components of the spread—i.e., the transmission of farming techniques versus the physical migration of farmers—has long been debated in the archaeological literature. Intriguingly, advances in genetics mean that quantitative assessments of these issues are now becoming possible, with many recent studies suggesting at least some migration of early farmers (e.g., [1,2]).

Recently there have been a number of studies using population dynamics models to describe the spread of neolithic farmers. While some recent work has focused on stochastic methods [3], most studies have built upon the pioneering work of Ammerman and Cavalli-Sforza [4] and sought the solution of a deterministic partial differential equation [5–7]. The success of this approach can be seen in a number of studies. For example, when using values for the model inputs determined theoretically or from the archaeological and anthropological literature, the authors of [8,9] found a reasonable agreement between the output of the numerical model and the large-scale features of the observed first-arrival times at neolithic sites, based on the radiocarbon dates of objects found at these sites. Unfortunately there may be many other parameter sets which provide equally well- if not better-fitting model outputs to the data. We seek these parameter sets by developing a rigorous statistical inference method to fit the model to the radiocarbon data.

In this paper we adopt a Bayesian approach to inference as this will also allow us to quantify parameter uncertainty in a

rigorous way. Additionally it allows us to quantify correlations between our model parameters and the global uncertainty in the fit of our model to the data. The authors are not aware of another study where these sophisticated statistical techniques have been used, allowing the determination not just of parameter estimates, but of a plausible range of parameter values, given by the posterior distribution.

The “wave of advance” is one of the most important concepts in modeling the spread of the neolithic, underlying the studies cited above; since the work of Ammerman and Cavalli-Sforza [4] and Clark [10], it has been widely accepted that the incipient farming spread from the Levant to western Europe in a systematic manner—an outward propagating “wave”—amenable to study by simple deterministic models. The rate of propagation of the wave is broadly constant throughout this spread, and various authors have estimated the speed of the wave front, U , from the radiocarbon data. In the present work, we try to produce a statistically reliable estimate of the wave speed U from the radiocarbon data. Unlike most if not all earlier studies, however, we explicitly account for the fact that U is a random variable, due both to the systematic and random errors inherent in the data, and to the inherent variability in the underlying population dynamics processes. The true spread is not well modeled on all scales, and at all locations, by a wave front advancing with a continuous speed; local variations do of course occur. Such local variations are not explicitly modeled within simple deterministic wave-front models, which effectively average out such small-scale variability and reproduce the spread well on the larger, continental scales.

As well as being the natural quantity to describe the spread, the wave speed has clear and important implications within most mathematical models of the spread. For example, within the Fisher-Kolmogorov-Petrovsky-Piskunov (FKPP) equation [11,12], most frequently used in models of demic diffusion, the front speed is directly linked to the population mobility (diffusivity) and growth rate. While the wave front has been most often studied within the context of the FKPP equation, many alternative models of population dynamics also predict waves of advance, with speeds dependent upon their various

^{*}a.w.baggaley@ncl.ac.uk

model parameters. For example, within multiple-population models related to the FKPP model but allowing for cultural conversion, the speed can be affected by the parameters controlling the conversion [13,14]; similarly, within other models of cultural transmission, it can be linked to the intensity and spatial range of contacts [15]. The same is true for many alternative models: e.g., models involving alternative parametrizations of growth processes and diffusion [5,16]; and models involving non-Laplacian diffusion (e.g., Lévy flights; [17]). Even within FKPP-like models with logistic growth, “time-delay” factors attempting to model the generational effects of population growth more realistically result in a modified relationship between the front speed and the basic demographic parameters [6,7]. The key quantity which must be inferred from the radiocarbon data is the wave speed, and so a wave-front-based approach such as that introduced here, which gives this quantity prominence, is a more natural model.

Edmonson [18] was the first to estimate the speed of the agropastoral transmission in Europe; he gave a value of 1.9 km/yr. Later, Ammerman and Cavalli-Sforza [4] gave a value of 1 km/yr, and most subsequent determinations of the speed of the neolithic wave front have also been of order $U \simeq 1$ km/yr [5,8]. There are notable regional variations, however. In particular, the Linearbandkeramik (LBK) culture spread along the Danube-Rhine corridor at a higher speed, perhaps as large as 5 km/yr [19,20]. The spread of the Impressed Ware ceramics along the Mediterranean coast has been estimated to be as fast as 10 km/yr [19], although lower estimates are also possible [21]. In contrast, there is no clear evidence for any significant acceleration along the northern and Atlantic coastline of Europe. In this paper we follow [8] in allowing for enhanced anisotropic spread along coastlines and along the courses of the Danube and Rhine rivers, in an attempt to model the above phenomena. We perform Bayesian inference for the magnitudes of these enhanced effects, and for the magnitude of the isotropic background spread; as a result, we are in a position to assess the extent to which these features of the model are genuinely required by the radiocarbon data.

Although the posterior distribution of the parameters of interest is analytically intractable, computationally intensive methods such as Markov chain Monte Carlo (MCMC) simulations can be used to generate samples from this distribution. The computational cost of simulating the wave of advance precludes the direct use of an MCMC scheme. We therefore approximate the arrival time of the wave front at each site using a Gaussian process emulator [22] as the computational speedup makes MCMC methods practicable.

The rest of this paper is organized as follows. We introduce our wave-front model, and compare its output to that obtained from a more traditional approach involving partial differential equations (PDEs), in Sec. II. In Sec. III we describe the data against which our model will be compared. We outline our statistical model, and the Bayesian inference scheme used to estimate our model parameters from the data, in Sec. IV. The results of our inference are described in Sec. V, and our conclusions are summarized in Sec. VI. Some technical details about our statistical methods, including the use of Gaussian process emulators, are presented as Appendixes.

II. MODELING THE PROPAGATING FRONT

The isotropic FKPP equation

$$\frac{\partial N}{\partial t} = \gamma N \left(1 - \frac{N}{K}\right) + \nabla \cdot (\nu \nabla N) \quad (1)$$

describes the evolution of population density $N(\mathbf{x}, t)$ at position $\mathbf{x}$ and time t , with the growth rate γ , diffusivity ν , and carrying capacity K as parameters. The solution to this equation forms a propagating wave front, which travels with a speed

$$U = 2\sqrt{\nu\gamma}, \quad (2)$$

which is dependent on both the diffusivity and the growth rate of the population, but importantly is independent of the carrying capacity. This result can be readily proved in one dimension [12], and also in two dimensions if the curvature of the front is sufficiently small (which is to be expected when the distance from the source is large compared with the front width) [23]. In a spatially heterogeneous environment (such that ν and/or γ vary with $\mathbf{x}$), the wave speed is clearly a function of position, $U(\mathbf{x})$.

Numerical solutions to the FKPP partial differential equation are most simply obtained by discretizing to a grid of points in space, using, e.g., finite-difference methods to approximate the spatial derivatives, and time-stepping the solution forward in time. Thus the local population density at each point on the grid is calculated at each time step. If we are focusing on the first-arrival time of the neolithic farmers at specific radiocarbon sites, however, then all modeling of the population behind the front is unnecessary; as indeed is the modeling of the equation ahead of the front, which solution of the FKPP equation also requires. A reasonable alternative is to model only the propagating front itself, which we do using a particle-based approach. This approach necessarily omits many complex demographic effects which may have occurred locally within the real spread; but that is also true of the FKPP equation most frequently used to model the neolithization. And given the limitations of the current data, it would be premature to adopt more complex demographic models for the spread on the continental scale; almost all models of the neolithization of Europe as a whole have therefore similarly focused on the propagation of the front.

We select a starting point for our initial population, and at a small radius from this point we approximate a circle with a small number of “particles” (points); these particles define our front. An alternative approach is to define a regional source; however, for simplicity here we take a localized source. We keep track of the index of the adjacent particles and so can easily define local tangent and normal vectors, and in particular the local unit outward normal $\hat{\mathbf{n}}$. (Here outward means in the direction of the advancing front.) We perform all simulations in spherical polar coordinates, at a fixed radius R , set to approximate the Earth’s surface ($R = 6378$ km), and define the position in terms of the polar and azimuthal angles ϕ and λ , respectively; i.e., $\mathbf{x} = (\phi, \lambda)$. Numerically, at each time step, we move the particle i , at position $\mathbf{x}_i = (\phi_i, \lambda_i)$ and with velocity $\mathbf{u}_i = U(\mathbf{x}_i)\hat{\mathbf{n}}_i$, a small amount in both the ϕ and λ directions according to

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{u}_i. \quad (3)$$

Here $\hat{\mathbf{n}}_i$ is the local outward normal, and t represents the time in years. As discussed in Sec. I, there are local deviations in the rate of spread of the neolithic, particularly along traversable waterways. We follow [8] in allowing an increased rate of spread along all coastlines, and also along the Danube-Rhine river systems. We label the enhanced coastal velocity as $\mathbf{V}_C$, and the river velocity as $\mathbf{V}_R$. In the partial differential equation approach, these velocities appear as additional advective terms in the FKPP equation [Eq. (1)], which can be identified with anisotropic diffusion [8]. In the wave-front approach, we can instead simply add this effect to the velocity experienced by particle i , which becomes

$$\mathbf{u}_i = U\hat{\mathbf{n}}_i + \mathbf{V}_i, \quad (4)$$

where the total advection from both river and coastal terms at position $\mathbf{x}_i$ is

$$\mathbf{V}_i = V_C \text{sgn}(\hat{\mathbf{n}}_i \cdot \tilde{\mathbf{V}}_{C,i}) \tilde{\mathbf{V}}_{C,i} + V_R \text{sgn}(\hat{\mathbf{n}}_i \cdot \tilde{\mathbf{V}}_{R,i}) \tilde{\mathbf{V}}_{R,i}. \quad (5)$$

Here $V_{C(R)}$ are the “amplitudes” for the river and coastal advective speeds, and $\tilde{\mathbf{V}}_{C(R),i}$ are normalized vectors in the direction of the relevant local advection (normalized to unit magnitude at points on the river or coast). The sign functions ensure that the sense of the advection (e.g., upriver or downriver) is that which enhances the outward speed of the locally expanding wave front. We use MCMC methods below to infer the acceptable range of the amplitude parameters V_C and V_R , given the radiocarbon data.

A. Spatial dependence

It is intuitively sensible that the local altitude should have a significant effect on the spread of the neolithic; and, indeed, early farming in Europe does not seem to have been practical at altitudes greater than 1 km above sea level (e.g., there are no LBK sites above this height in the Alpine foreland [24]). There is also significant evidence [4,25] that the latitude (acting partly as a proxy for the climate) had a significant impact on the productivity of the land, and therefore the ability to farm. Another latitudinal effect noted in the literature is an increased competition in the north with the preexisting mesolithic population [7]. Both of these factors motivate a decreased wave speed at higher latitudes.

In order to introduce these spatial dependences into the model, we use arrays of geographical altitude data taken from the ETOPO1 1-min Global Relief database [26], taking a data set with spatial resolution of 4 arc min. This forms a 740×1100 mesh, with approximate longitudinal boundaries of 15° W and 60° E and latitudinal boundaries of 25° N and 75° N.

At points on this grid, we calculate the local velocity to reflect the factors outlined above. On land, we require the speed to decrease to zero at altitudes above 1 km, using a smooth approximation to a step function. To allow for limited sea travel we use a different form of cutoff at low altitude, setting the speed of the wave front to decrease exponentially with the distance to the nearest land (d_c). We also include a linear dependence of U on latitude. As a result, we calculate

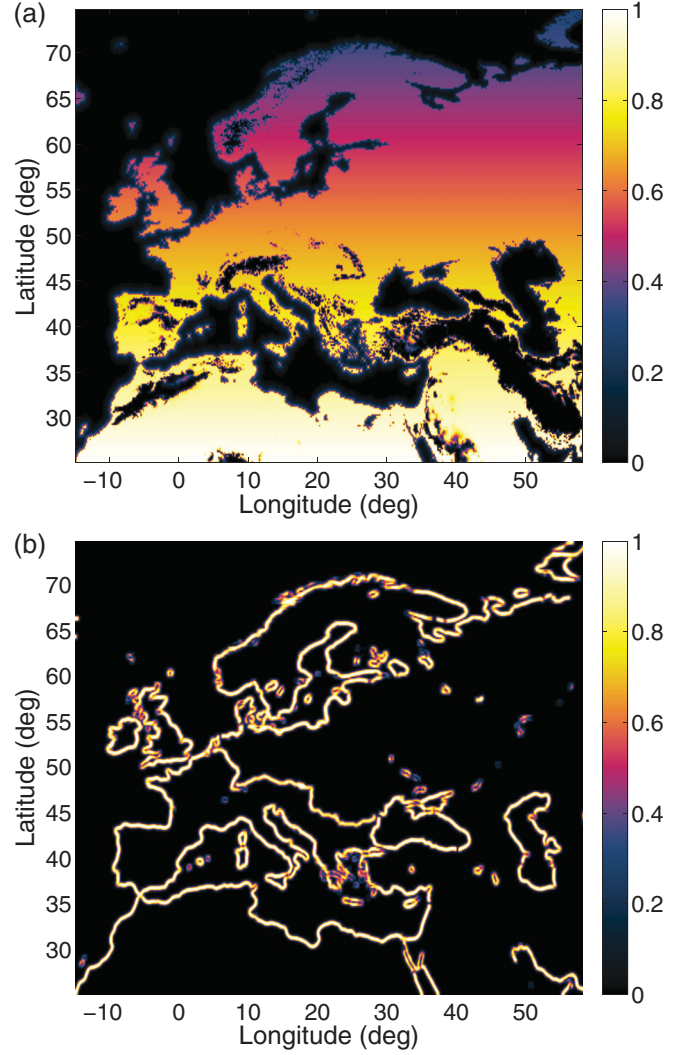


FIG. 1. (Color online) (a) Spatial variation of the speed $U(\mathbf{x})$ [from Eq. (6)]; (b) magnitude of the distance-weighted coastal and river vectors $|\tilde{\mathbf{V}}_i|$ [as in Eq. (5)].

the local velocity on our grid as

$$U = U_0 \left(\frac{5}{4} - \frac{\phi}{100^\circ} \right) \begin{cases} \frac{1}{2} - \frac{1}{2} \tanh\{10(a - 1 \text{ km})\}, & a > 0, \\ \exp(-d_c/10 \text{ km}), & a < 0. \end{cases} \quad (6)$$

Here U_0 is the background amplitude determining the mean rate of spread; we expect U_0 to be of order 1 km/yr, but use our MCMC methods below to infer the acceptable range of this parameter (given the radiocarbon data) more rigorously, along with the parameters V_C and V_R introduced above. The spatial variations in U are illustrated in Fig. 1 (top).

To deal with the advection terms, the river and coastline vectors used in this study are taken from [27], which contains vector data of the world’s coastlines and major rivers. We take a subset of these data, which contains all the coastlines within our domain, and the river vectors corresponding to the Danube and Rhine. To obtain values of $\mathbf{V}_{C(R)}$ at each point on our mesh, a distance-weighted contribution is taken from each of the irregularly spaced vector data segments which

define the river in [27]. Specifically, the contribution from each segment is weighted by $\exp(-d_{\text{vec}}/15 \text{ km})$, where d_{vec} is the distance between the grid point and the river or coastal vector (in kilometers). The magnitudes of the river and coastal vectors are shown in Fig. 1 (bottom). The small-magnitude features visible in some inland nonriver locations in this plot are associated with small lakes, which are included in the coastline database. These features have no significant effect on our model.

It is perhaps worth commenting on our use of present-day information about coastline and river locations, since these locations have changed over the time scale of the spread we are modeling; most obviously, coastlines have changed as a result of sea-level changes arising principally from postglacial isostatic adjustment. During earlier work based on the PDE approach, we investigated the effect of such changes, using a sea-level model supplied by geophysicists from Durham University (e.g., [28–30]). The effect on our model was negligible, since the most significant changes in sea level were in the north, and had largely occurred by the time that the neolithic wave reached the northern coasts (so that potentially important land bridges had already disappeared) [31]. We have not tried to account for changes in the courses of rivers, but we do not expect that such relatively local changes would have a significant effect on the large-scale spread on which we focus.

In our wave-front model, to calculate the local speed U appropriate at the precise position of particle i [as is needed for Eq. (4)], we use bilinear interpolation from the values at the four closest mesh points. Similarly, the local advective vectors at the precise position of particle i are also obtained using bilinear interpolation from the closest mesh points; it is these local vectors which appear in Eq. (5).

B. Wave-front algorithms

We monitor the separation between the particles, and if this becomes larger than some specified value δ , we introduce a new particle in order to maintain a roughly constant resolution along the front; see Fig. 2(a) for an illustration of this process. Due to the irregular nature of the velocity map shown in Fig. 1, the wave front can separate around low-velocity regions (e.g., mountain ranges), and subsequently remerge. We apply algorithms initially used to model the evolution of magnetic flux tubes in astrophysical simulations [32] to merge wave

fronts in the particle model. At every time step we check the distance between each particle and all the other particles in the simulation. If the separation of any two particles (which are not neighbors) is less than the resolution length δ , then we remove the encroaching points and switch the ordering of the loops, so as to merge the fronts; a schematic of this process is shown in Fig. 2(b). Typically this process results in a merging of the main front and the creation of a small loop behind the main front, which we normally remove from the simulation to avoid unnecessary numerical effort. Snapshots from a simulation where these small loops are not removed are shown in Fig. 3. In the work reported here, we take δ to be equal to our grid spacing of 4 arc min.

C. Comparison between simulations

We now present qualitative results on the comparison between the wave-front model and the PDE model for a typical parameter set. For these (and subsequent) calculations, we place the initial source of the farming population at 40° N , 35° E , within the Fertile Crescent. The starting time for the spread is taken as 6572 yr cal BC as this date is consistent with [8] and the radiocarbon data at the site Tell Kashkashok. The FKPP equation (1) is solved numerically, approximating spatial derivatives using a second-order finite-difference scheme, and time stepping using an Euler scheme. The spatial resolution is 4 arc sec, on a 740×1100 mesh. This is the same mesh introduced above to control the spatial variations for our wave-front model; the same spatial variations are used for the FKPP equation. The reduced computational effort of the wave-front method provides us with a significant speedup, with a typical simulation taking approximately 10 s on a single processor with a clock speed of 2.67 GHz. The corresponding solution to the FKPP equation, with a time step satisfying the Courant-Friedrichs-Lewy condition [33], requires approximately 24 h on an eight-processor cluster with hyperthreaded Intel Xeon quadcore processors of the same clock speed.

While we do not expect an exact agreement between the two models, we do expect their arrival time at a particular site to be similar. Indeed, we do find a reasonable agreement between the two models, with a maximum discrepancy in the arrival time of approximately 120 yr and typical values less than 50 yr

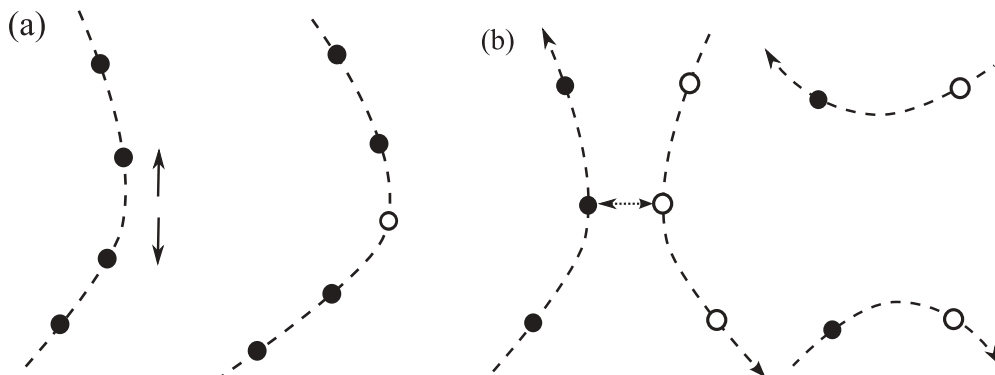


FIG. 2. Schematic of the algorithms: (a) for the insertion of a new point (open circle) in a spreading front; (b) for the removal of encroaching points (central circles) in merging fronts.

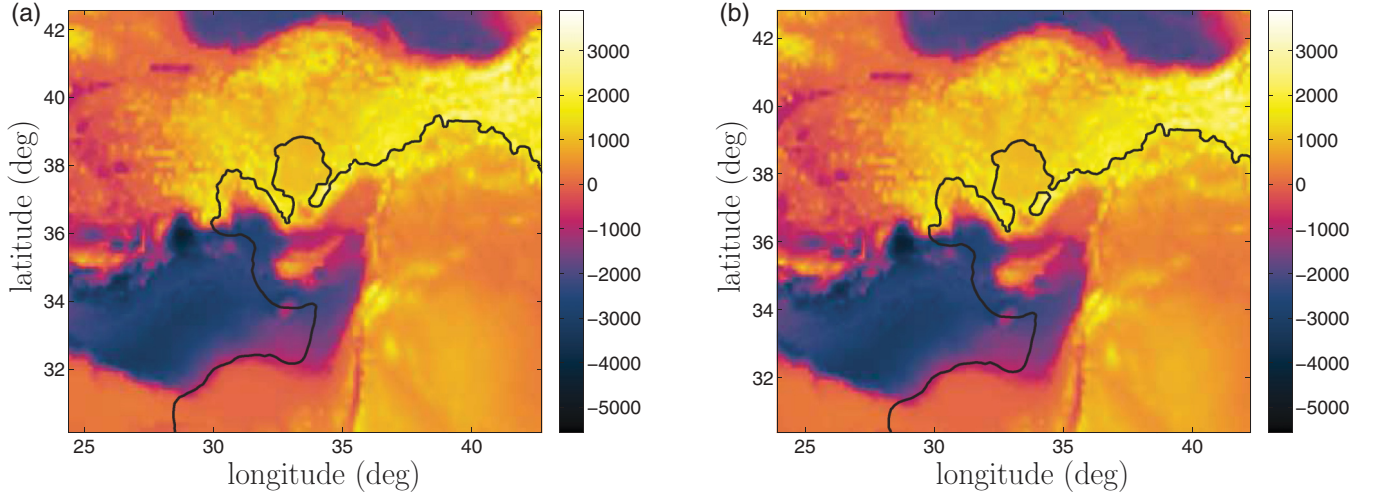


FIG. 3. (Color online) The wave-front model plotted just before (a) and after (b) a merging event. The color gradient plot displays the altitude (in meters) for this region (the eastern Mediterranean). Note that the small loop shown behind the merged front would normally be removed from the calculation, but has been left in here for illustrative purposes.

(for a simulation covering of order 5000 yr). Figure 4 shows snapshots at four times during the simulations.

III. RADIOCARBON DATA

We use a compilation of 302 dates from sites in southern and western Europe from [25,34,35]. These data contain multiple dates per site and so we determine a single date for each site by using a method based on [20]. The method we use is described in detail by Davison *et al.* [36]. Briefly, for sites with at least eight date measurements, a χ^2 statistical test is used to determine the most likely first-arrival date from a coeval subsample, and for sites with fewer measurements, we use a weighted mean of these measurements. Figure 5 shows a plot of the radiocarbon sites shaded according to the estimated first-arrival time t_i obtained from this statistical treatment.

In this paper we focus on a statistical model with a single error term accounting for mismatch between the data and the wave front; this is described in the next section. While this approach is not entirely satisfactory, the additional complication of properly accounting for varying site errors would add another level of complexity, which was deemed prohibitive for this initial investigation. However, a statistical model which accounts for errors in the dates that can vary between sites may be adopted in future studies.

IV. BAYESIAN INFERENCE

We now outline the proposed statistical model and provide details of our Bayesian inference scheme. Let $\tau(\mathbf{x}_i|\boldsymbol{\theta})$ denote the time at which the wave front arrives at site i , at position $\mathbf{x}_i$, for $i = 1, \dots, n$, where $n = 302$ is the number of radiocarbon sites in our data set. Here $\boldsymbol{\theta} = (U_0, V_C, V_R)^T$ is the vector of model parameters about which we want to draw inferences. The observed arrival time at site i is denoted by t_i and these times are displayed in Fig. 5. Our statistical model assumes that these data are generated by the wave-front model subject to (spatially) independent normal errors. Specifically, at site i

we have

$$t_i = \tau(\mathbf{x}_i|\boldsymbol{\theta}) + \sigma\epsilon_i, \quad i = 1, \dots, n, \quad (7)$$

where the ϵ_i are independent and identically distributed standard normal random variables and σ is the spatially homogeneous standard deviation, allowing for a mismatch between the model and the observations (i.e., local variations in the arrival of the wave front, corresponding to the expected local deviations from the idealized, global model).

By adopting a Bayesian approach to the problem of inferring the model parameters, we express initial beliefs about likely parameter values via a prior distribution, denoted $\pi(\boldsymbol{\theta}, \sigma)$. We then construct the posterior distribution of our parameters, given the observed arrival times. Bayes' theorem gives this posterior distribution as

$$\pi(\boldsymbol{\theta}, \sigma | \mathbf{t}) \propto \pi(\boldsymbol{\theta}, \sigma) \pi(\mathbf{t} | \boldsymbol{\theta}, \sigma), \quad (8)$$

where $\pi(\mathbf{t} | \boldsymbol{\theta}, \sigma)$ is the likelihood function, i.e., the joint probability of the observed arrival times, regarded as a function of the parameter values. If we assume the model outlined in Eq. (7), then we can write the likelihood function as

$$\pi(\mathbf{t} | \boldsymbol{\theta}, \sigma) \propto \sigma^{-n} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n \{t_i - \tau(\mathbf{x}_i|\boldsymbol{\theta})\}^2 \right]. \quad (9)$$

We specify our fairly weak *a priori* beliefs about $\boldsymbol{\theta}$ by adopting independent log-normal distributions for the components U_0 , V_C , and V_R , with modes chosen to match previous archaeological estimates [4,21,25]. Specifically we take $U_0 \sim \text{LN}(0.5, 0.71^2)$, $V_C \sim \text{LN}(1, 0.5^2)$, and $V_R \sim \text{LN}(2.2, 0.8^2)$. We use a weakly informative inverse Gamma prior for the global error parameter, with $\sigma^2 \sim \text{IG}(5, 10^6)$.

Due to the complex dependence of the wave-front solution on the model parameters, the posterior in Eq. (8) is analytically intractable. Markov chain Monte Carlo methods are commonly used, in the context of Bayesian inference, to sample intractable posterior distributions. These methods aim to construct a Markov chain whose invariant distribution is the desired posterior distribution. Such approaches are particularly

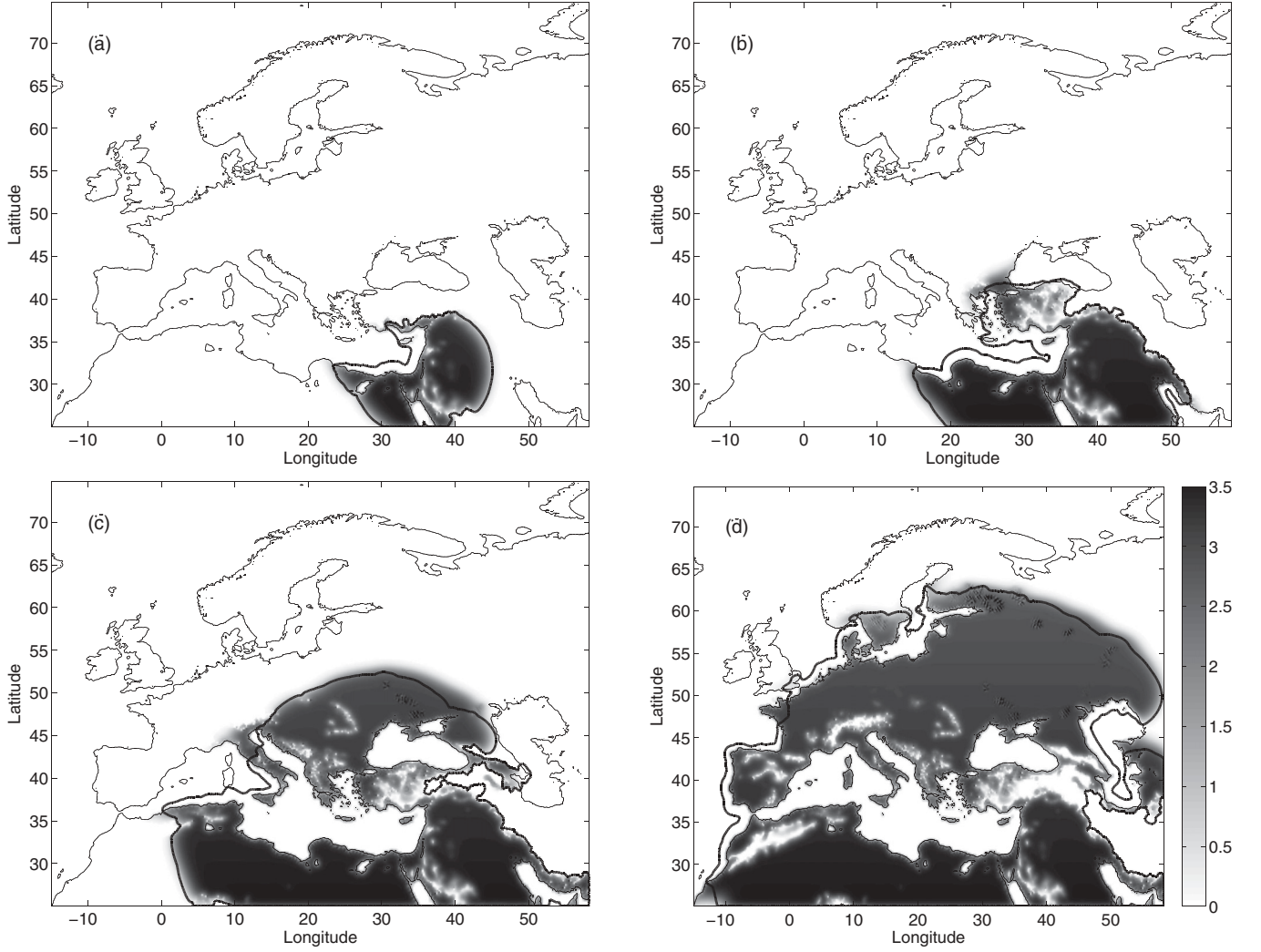


FIG. 4. A comparison between the numerical solution to the FKPP equation and our propagating wave-front model. The output from the wave-front model is shown by the thick black lines. The corresponding output from the FKPP model is shown by the grayscale plots of the population density, with the scale given to the right of panel (d); the values are in terms of people per square kilometer. The four panels are for times 1000 (a), 1800 (b), 3000 (c), and 4600 (d) yr after the start of the simulations. In the wave-front simulations $U_0 = 1$ km/yr, $V_C = 2$ km/yr, and $V_R = 0$. A consistent parameter set is used in the FKPP simulation.

useful for Bayesian inference since the target distribution need only be known up to proportionality. A recent review of these methods can be found in [37].

In this paper we focus on a Gibbs sampler [38]. This particular MCMC scheme can be useful for sampling from high-dimensional distributions, and requires the ability to sample from the full conditional distribution of each parameter (or, more generally, subsets of parameter components). In the absence of analytically tractable full conditionals, a Metropolis-Hastings scheme can be used for this. Such an approach is often termed *Metropolis within Gibbs*, and its use is outlined in the next section.

A. Markov chain Monte Carlo algorithm

In this section we provide details pertinent to our implementation of the MCMC scheme. A more detailed description of the algorithm can be found in Appendix A.

We consider a Gibbs sampling strategy where we alternate between draws of θ and draws of σ^2 (and therefore σ) from their full conditional distributions. The form of the statistical model and its inverse Γ prior permit an analytically tractable full conditional for σ^2 . Consequently realizations of σ can be sampled directly. The full conditional density of θ , namely, $\pi(\theta|\sigma, t)$, however, is intractable and we therefore use a Metropolis-Hastings scheme to sample from the corresponding distribution. In brief, a Markov chain is constructed by generating candidate values of each component of θ via a symmetric random walk with normal innovations on a logarithmic scale: this ensures that proposed parameter draws are non-negative. A proposed value is accepted as the next value in the chain with a probability that ensures the Markov chain has an invariant distribution given by the distribution of interest. If a proposal is not accepted then the next value for that parameter is taken to be its current value. The acceptance probability requires that the target density can be evaluated up to proportionality. Each MCMC iteration therefore requires a

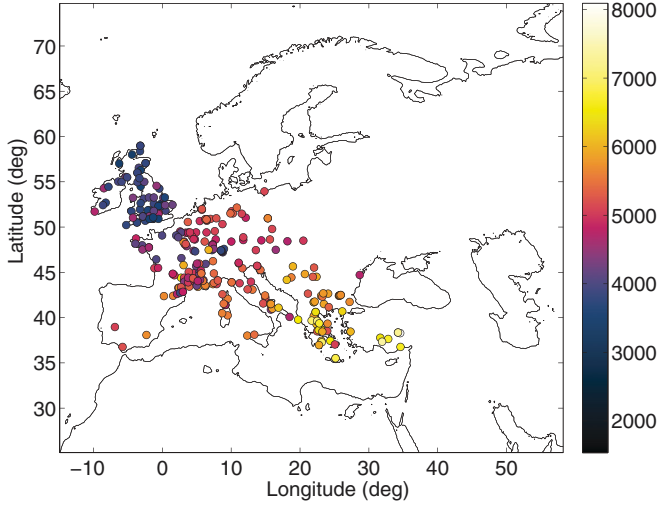


FIG. 5. (Color online) The radiocarbon sites we use to constrain our simulations. The color bar shows the estimated first-arrival time at a site, in time BC, from the radiometric data.

single run of the wave-front model expanding across the whole of Europe.

Unfortunately, the number of MCMC iterations required to produce near independent draws from the joint posterior distribution precludes using the wave-front model to evaluate each $\tau(\mathbf{x}_i|\boldsymbol{\theta})$; the individual wave-front model calculations run too slowly, given the large number of iterations required.

To proceed, we seek a faster approximation of the first-arrival times from the wave-front model. One option is to use a deterministic approximation, such as linear interpolation or cubic splines. Initially the wave-front model would be run at a specific set of parameter values, and the arrival time at each site stored. Parameters within the interpolation method could readily be computed from this output. The arrival time at new parameter sets could then be approximated using the chosen deterministic “emulator.” In the statistical literature [39], Gaussian process emulation is favored, as this not only interpolates smoothly between design points but also quantifies levels of uncertainty around interpolated values. Further details on how to build and test such emulators can be found in Appendix B. Using these emulators to approximate the first arrival times $\tau(\mathbf{x}_i|\boldsymbol{\theta})$ at each site makes the MCMC scheme outlined above

computationally practicable. The scheme produces a sample from the joint posterior distribution of our model parameters.

V. RESULTS

We now present results obtained from the output of the MCMC scheme. We performed 5.5×10^6 iterations of the algorithm before discarding the first 5×10^5 parameter draws as “burn in” to allow the chain to converge. The remaining 5×10^6 iterates were then thinned to reduce the autocorrelation in the sample: we took every 500th iterate, leaving a sample of 10^4 (almost) uncorrelated values from the joint posterior distribution. We assessed convergence of the MCMC scheme by repeating the above procedure for many different starting parameter sets (randomly drawn from the prior distribution) and found no problem with convergence.

The output of the MCMC scheme is summarized in Fig. 6. It shows kernel density estimates [40] of the marginal posterior probability density function, for each of U_0 , V_C , V_R , and σ . For the three parameters in our mathematical model, the modes of the marginal posterior distributions (in black in Fig. 6) are of the magnitude expected from other studies in the literature (as described in Sec. I). Compared with their respective prior distributions (in red), the posterior distributions are considerably tighter, showing that the radiocarbon data have indeed been informative, and have effectively constrained the plausible range of model parameters. For example, posterior samples of the background wave speed U_0 (with a mode of approximately 1 km/yr, and a 95% range of 0.79–1.41 km/yr), are in good agreement with the studies cited in Sec. I [4,5,8]. In terms of an FKPP model, with a growth rate of $\gamma \simeq 0.02 \text{ yr}^{-1}$ (of the order typically used in such models), this would correspond to a diffusivity of $\nu \simeq 13 \text{ km}^2/\text{yr}$; this value is also comparable to those typically used in FKPP models.

The only previous studies which have modeled an enhanced population mobility along waterways [8,9] were motivated by results based on specific local phenomena. For example, the incorporation of enhanced coastal speeds within the model was motivated by radiocarbon evidence for the spread of the Impressed Ware culture along the coastline of the western Mediterranean, with some estimates of speeds of order 10 km/yr [19,21]. While some form of enhanced spread along the coastline of this region may be required, extrapolating to a similar spread along all of Europe’s coasts might very well give an inferior fit to the data as a whole. For example, if

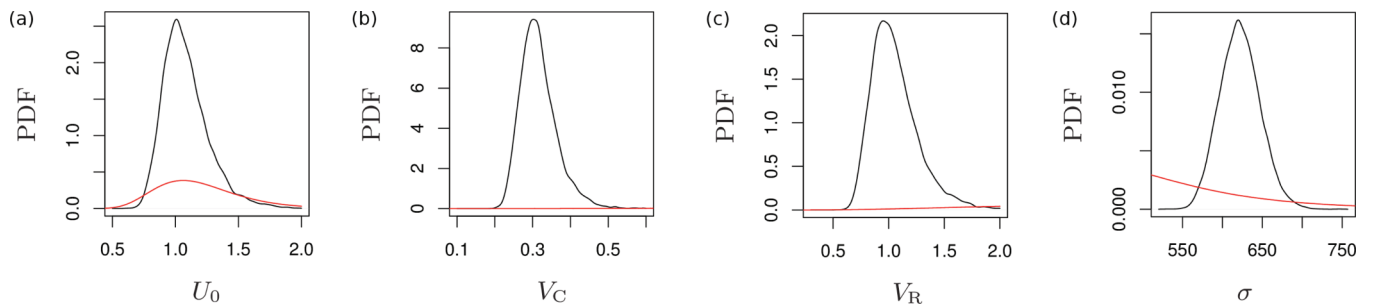


FIG. 6. (Color online) Marginal posterior densities (upper black lines; prior distribution, lower red line) for (a) U_0 , (b) V_C , (c) V_R , and (d) σ , based on the (thinned) output of the MCMC scheme, using a Metropolis within Gibbs sampler. PDF represents the probability density function.

we take the rate of spread along all of Europe's coastlines to be of order 10 km/yr, then the wave front may arrive at many sites far earlier than the radiocarbon data suggest. The inference presented in this paper seems to confirm this, as marginal posterior samples of the global coastal propagation speed V_C (with a mode of around 0.3 km/yr, and a 95% range of 0.23–0.41 km/yr) are markedly lower than the values quoted for the Impressed Ware culture.

Posterior samples of V_C are also significantly lower than the corresponding value used in [8] (20 km/yr). It may be that, compared with that earlier work, our more rigorous method of comparing models against the data suggests an improved model fit without a large enhanced coastal speed, and with regional data variations (such as those associated with the Impressed Ware culture) largely accounted for within the global error parameter in our statistical model (discussed further below). This may not entirely explain the difference between the two studies, however; it is also possible that the implementation of coastal advection in [8] has exaggerated the overall magnitude of this effect. The value quoted there for the advection actually only applies at points exactly on the coast, whereas nearby locations experience a reduced velocity, decreasing with their distance from the coast; as a result, the mean, effective, advective speed may be somewhat lower than the peak value quoted.

In terms of the river advection, the marginal posterior values of V_R are clearly non-negligible (with a mode of approximately 1.0 km/yr, and a 95% range of 0.72–1.38 km/yr); it is comparable to the background spread modeled by U_0 , vindicating the suggestion of anisotropic spread along these river basins. However, this value is significantly smaller than the values normally quoted for the spread of the LBK culture (of order 5 km/yr) [8]. This deviation may be due to the reasons discussed above for the coastal velocity. However, the discrepancy with the widely accepted archaeological time scale for the LBK culture (which does not refer to a spatial mathematical model, but is obtained directly from a coeval set of radiocarbon dates) requires further explanation. Looking in detail at the data in this region, it may be that the earliest dates associated with the LBK culture simply do not correspond well to spread by a continuous wave of advance

(anisotropic or otherwise); rather, early settlements may have been formed by something like a leapfrog or pioneer mechanism, and the whole region only settled (and outward spread continued) after some subsequent delay. In this paper we have studied a model of large-scale spread, but it may be the model is too crude and would need small-scale refinement to explain the spread of the LBK culture.

For the global error parameter in our statistical model [Eq. (7)], σ , the marginal posterior distribution, is centered on a value of order 600 yr (see Fig. 6). The 95% confidence range is 577–671 yr. This is significantly larger than the values typically quoted for the effective minimum uncertainty of radiocarbon dates for this period, of order 160 yr [20]. The latter value is derived empirically from well-explored, archaeologically homogeneous sites, effectively allowing for sample contamination and other sources of errors; this may be contrasted with the quoted laboratory uncertainties, which characterize only the accuracy of the laboratory measurement, regardless of the provenance of the sample. The global parameter (σ) in our model, however, does not merely reflect the uncertainty in the dating, but also allows for the mismatch between our mathematical model of the spread (a globally continuous wave of advance) and the regional variations present in the actual spread. Thus our inference suggests that a simple global wave of advance across southern and western Europe, while remaining a good model on the continental scale, should be considered a good model only on time scales of order 600 yr (and thus length scales of order 600 km) or greater; on shorter time scales (and length scales), significant local variations should be expected. As noted briefly above (in our discussion of the advective velocities), this parameter within our model may to some extent allow for regional variations that might alternatively be modeled by specific regional effects (e.g., river advection), thus explaining the relatively low values of our inferred advective speeds. The posterior distributions suggest that this type of fit—requiring relatively large global uncertainty, but then favoring relatively low local advective speeds—is the optimum way of explaining our data set of first-arrival times within a wave of advance model of the sort presented here. Of course, the conclusions of this inference may depend upon specific features of the

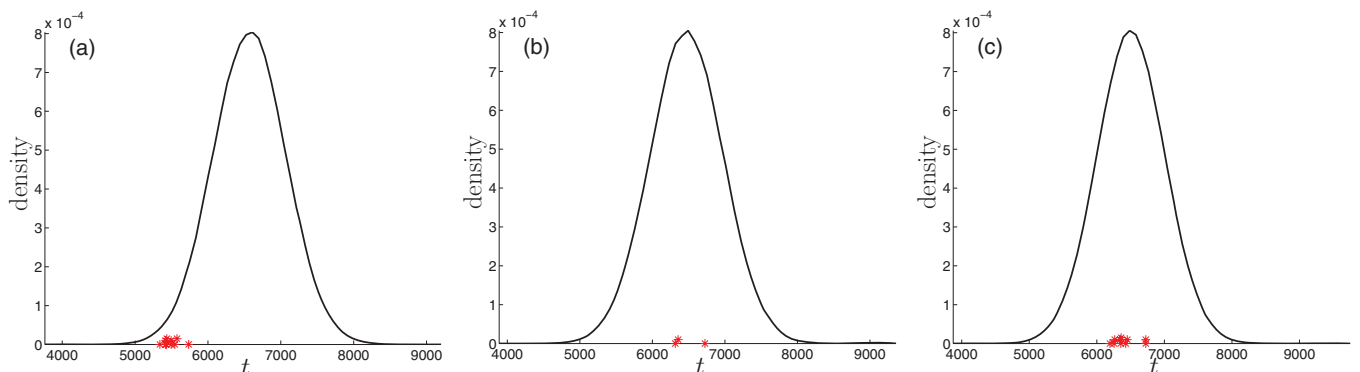


FIG. 7. (Color online) Predictive densities for the arrival time at three sites plotted as black lines, with the observed radiocarbon dates plotted as (red) asterisks. (a) Site Kremenik, located at 42.3° N, 23.27° E, an example of a bad model fit, where our model predicts a much earlier arrival time than is presently observed; this point can be plotted as a triangle in Fig. 8. The middle and right panels are examples of more typical agreement between the model and data. (b) Agrissa Magoula (39.63° N, 22.47° E) and (c) Seskto (39.28° N, 22.82° E).

models used here (both mathematical and statistical), and may vary for different models. In extensions to the current work we will explore alternative models, e.g., allowing for increased regional variation in the coastal advection, and allowing for spatial correlation between nearby sites within the statistical model. We will also consider additional, more recent, radiocarbon data, which will provide observed arrival times at new sites, and will thereby allow an out-of-sample assessment of prediction error.

In addition to performing inference for the model parameters, we use predictive simulations to assess the validity of the statistical model and the underlying model of the wave of advance. The posterior predictive distribution of the arrival time at a site i , $t_{i,\text{pred}}$, can be determined as follows. Using Eq. (7) we have that

$$t_{i,\text{pred}}|\theta, \sigma \sim N(\tau(\mathbf{x}_i|\theta), \sigma^2),$$

where $\tau(\mathbf{x}_i|\theta)$ is the arrival time of the wave front, approximated by the emulator. We therefore take each sampled parameter value $(\theta^{(j)}, \sigma^{(j)})$ from the MCMC output and generate a realization from the predictive arrival time distribution by simulating realizations from $t_{i,\text{pred}}^{(j)}|\theta^{(j)}, \sigma^{(j)}$. We thus obtain a sample of first-arrival times at each site. Figure 7 shows the predictive densities for three sites.

To see where our model is failing to agree with the radiocarbon data, sites where the observed radiocarbon date falls outside an approximate 95% credible interval for predicted first arrival are plotted as filled symbols in Fig. 8; sites which fall inside this interval are plotted as open circles. We distinguish between sites where our model predicts an earlier arrival time than is observed in the radiocarbon data, and those where our model arrives late. Where we predict an earlier arrival time, it is quite possible that the radiocarbon data at the site are simply from a relatively late settlement within this local region, and earlier data there have yet to be discovered. Where the model predicts a later first arrival time than is observed, then this may

be an indication that some localized process, which we have not included in our model, has caused a much faster spread in this region.

VI. CONCLUSIONS

In this paper we have introduced an innovative wave-front model, which allows the efficient simulation of the spread of a wave of advance model (with both isotropic and localized anisotropic components of spread); we have applied this model to the spread of neolithic culture across Europe (with the localized anisotropy being associated with a hypothesized enhanced rate of spread along certain waterways). We adopted a Bayesian approach to the problem of inferring the model parameters given observed arrival times, which we assumed were given by the wave-front model but subject to Gaussian error. A Markov chain Monte Carlo scheme was used to sample the intractable posterior distribution of the model parameters. To alleviate computational cost, we constructed Gaussian process emulators for the arrival time of the wave front at each radiocarbon site. As a result, we obtain the marginal posterior probability distributions for the model parameters of interest: the background rate of spread (U_0), and the enhanced rates of spread associated with coastlines (V_C) and with the Danube-Rhine river systems (V_R). To our knowledge, this is the first attempt to apply such inference techniques to this problem.

We find that the posterior variance is reduced (relative to the prior variance), suggesting that the data have been informative. Marginal posterior samples of U_0 , with a modal value of order 1 km/yr, are consistent with previous studies [5,8]. Modal values for V_C and V_R are of order 0.3 km/yr and 1 km/yr, respectively. This value for the river advection ($V_R \simeq 1$ km/yr) is clearly comparable to the speed of the background spread (U_0), confirming that an enhanced spread within these river basins can be robustly concluded from the data. This value is nevertheless significantly smaller than the value of 5 km/yr often quoted for the rate of spread of the local neolithic culture (the LBK culture) [19,20]. A closer inspection of the relevant data suggests that the spread of this particular culture may not be particularly well modeled by a continuous wave of advance, and subsequent models for this region may wish to pursue other possibilities; the estimate of 1 km/yr given above should simply be considered as the best-fitting value within the constraints of the current model. The relatively low modal value for the coastal advection ($V_C \simeq 0.3$ km/yr) suggests that such an advection, while not negligible, should not be considered particularly significant throughout Europe as a whole. This is perhaps not surprising, given that the principal motivation for this effect applies only to a specific region of Europe (the western Mediterranean coastline, along which the Impressed Ware culture spread [19,21]).

In addition to performing inference for the parameters characterizing the wave front, we also infer the global error σ (here formally introduced within our statistical model), representing both uncertainty in the radiocarbon dates and the misfit between our simple global wave-front model and the true spread (with its regional variations and local anomalies). The posterior modal and mean values for σ are of the order of 600 yr, significantly larger than the uncertainty normally

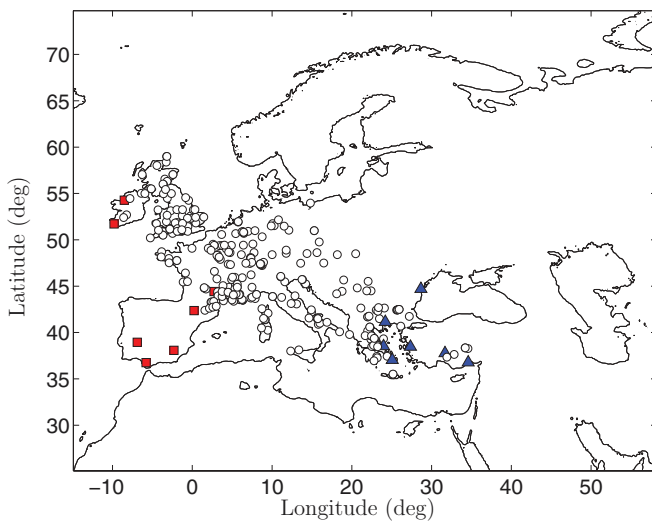


FIG. 8. (Color online) Discrepancies between the predicted first-arrival time (modal values of $t_{i,\text{pred}}$) and the observed values (t_i). Filled blue triangles (red squares) show sites where the model predicts anomalously early (late) arrivals. Open circles show sites with more acceptable agreement.

associated with radiocarbon dates for sites of this period (of order 160 yr) [20]. We therefore argue that this time scale, of order 600 yr (and consequently also a length scale, of order 600 km), is the scale at which the spread of the neolithic in Europe can be considered well modeled by a simple wave of advance: at longer time scales and length scales (and clearly on the continental scale), such a model of the spread performs well; at shorter time scales and length scales, significant local deviations from such a simple spread must be expected. The quantification of this scale is an important result from our inference.

Of course, the conclusions above must depend to some extent upon the specific models introduced here (both our mathematical wave-front model and the statistical model involving a global error parameter). In extensions to this work, we intend to investigate the robustness of these results with respect to various changes in these models. In our wave-front model, we first intend to investigate the possible importance of more regional variations within the enhanced spread along waterways. For example, we plan to allow different amplitudes of coastal enhancements within different regions, thus allowing us to explore more effectively the possibility of a regionally enhanced spread in the western Mediterranean, as proposed for the Impressed Ware culture there. We may also allow for advective velocities along other river systems, in addition to the Danube and Rhine.

Implicit in our statistical model is the assumption of spatially homogeneous normal errors. This assumption may be unnecessarily restrictive, and alternative statistical models (with more complex error structures) should be considered. For example, the statistical model may be adapted to allow for dates at different sites having differing uncertainties; building this into the model might result in a more meaningful fit (e.g., avoiding the possibility that a single global error parameter, as used here, may be unnecessarily smoothing out the fit everywhere). Further model refinements may also be possible. The wave of advance clearly expects that nearby sites will have similar arrival times; in the current model, however, nearby sites are not linked in any way. We will therefore allow for spatial correlation between nearby sites, potentially helping to smooth out locally anomalous dates, and also allowing another estimate of the scales over which the radiocarbon data correspond well to a simple wave of advance.

The neolithization of Europe is obviously not the only possible application for the methods introduced here, and applications to other regions or to other prehistoric periods (e.g., the dispersal of paleolithic cultures) also have great potential. Other applications would of course have their own difficulties, with one likely challenge being the relative scarcity of empirical data in many cases. One such case is the spread of neolithic culture from the Near East to South Asia; there are significant gaps in the radiocarbon record between these regions, and it would be of great interest to see how our methods could help to model this spread.

APPENDIX A: THE METROPOLIS-HASTINGS ALGORITHM

We provide a detailed step-by-step description of the MCMC scheme we use to sample from the posterior distribu-

tion of the model parameters, $\pi(\boldsymbol{\theta}, \sigma | \mathbf{t})$. (This type of scheme is well established within the statistical literature [37,38], but is presented here to help the more general readership to appreciate the current work.)

We use a Gibbs sampling strategy, alternating between draws of the full conditional distributions $\pi(\sigma | \boldsymbol{\theta}, \mathbf{t})$ and $\pi(\boldsymbol{\theta} | \sigma, \mathbf{t})$. Algorithmically, we perform the following steps:

- (1) Initialize $\sigma^{(0)}$ and $\boldsymbol{\theta}^{(0)}$. Set $j = 1$.
- (2) Draw $\sigma^{(j)} \sim \pi(\cdot | \boldsymbol{\theta}^{(j-1)}, \mathbf{t})$.
- (3) Draw $\boldsymbol{\theta}^{(j)} \sim \pi(\cdot | \sigma^{(j)}, \mathbf{t})$.
- (4) Set $j := j + 1$ and go to step 2.

The resulting Markov chain has an invariant distribution given by $\pi(\boldsymbol{\theta}, \sigma | \mathbf{t})$ [37]. The full conditional for σ can be sampled straightforwardly as, if $\zeta = \sigma^{-2}$, then

$$\zeta | \boldsymbol{\theta}, \mathbf{t} \sim \Gamma(A, B), \quad (\text{A1})$$

where

$$A = a + \frac{n}{2}, \quad B = b + \sum_{i=1}^n \{t_i - \tau(\mathbf{x}_i | \boldsymbol{\theta})\}^2 / 2.$$

Hence, in step 2 of the Gibbs sampler, $\sigma^{(j)}$ is generated by first drawing $\zeta^{(j)} | \boldsymbol{\theta}^{(j-1)}, \mathbf{t}$ and then setting $\sigma^{(j)} = 1/\sqrt{\zeta^{(j)}}$. Since the full conditional for $\boldsymbol{\theta}$ is analytically intractable we use a Metropolis-Hastings update in step 3. Define

$$\boldsymbol{\lambda} \equiv (\lambda_1, \lambda_2, \lambda_3)^T = (\log(U_0), \log(V_C), \log(V_R))^T$$

and note that under the prior specification adopted for $\boldsymbol{\theta}$, each component λ_i , $i = 1, 2, 3$, follows a normal distribution (independently) *a priori*. In step 3 of the Gibbs sampler we propose a new value $\boldsymbol{\lambda}^*$ via a symmetric random walk with normal innovations, that is,

$$\lambda_i^* = \lambda_i + \omega_i, \quad \omega_i \sim N(0, \delta_i^2), \quad i = 1, 2, 3,$$

where the δ_i are tuning parameters, the choice of which will influence the mixing of the Markov chain. Large values of δ_i will lead to small acceptance probabilities, and the chain will rarely move; whereas small δ_i will lead to many accepted proposed values, but slow exploration of the parameter space. We accept the proposed value and take $\boldsymbol{\lambda}^{(j)} = \boldsymbol{\lambda}^*$ with probability α ; otherwise we take the current value $\boldsymbol{\lambda}^{(j)} = \boldsymbol{\lambda}^{(j-1)}$. The acceptance probability is given by

$$\alpha = \min \left\{ 1, \frac{\pi(\boldsymbol{\lambda}^*) \pi(\mathbf{t} | \boldsymbol{\lambda}^*, \sigma^{(j)})}{\pi(\boldsymbol{\lambda}^{(j-1)}) \pi(\mathbf{t} | \boldsymbol{\lambda}^{(j-1)}, \sigma^{(j)})} \right\}, \quad (\text{A2})$$

where $\pi(\boldsymbol{\lambda})$ denotes the prior density ascribed to $\boldsymbol{\lambda}$ and $\pi(\mathbf{t} | \boldsymbol{\lambda}, \sigma)$ is given by Eq. (9) with $\boldsymbol{\theta} = \exp(\boldsymbol{\lambda})$.

APPENDIX B: EMULATION

The MCMC inference scheme typically requires many iterations, with each iteration requiring a full simulation of the expanding neolithic front to evaluate the likelihood function. As simulations of the front are computationally expensive, we emulate the model using Gaussian processes (GPs) [22], that is, stochastic approximations to the arrival times obtained from the wave-front model. These methods are widely used in the computer models literature; see, for example, [39] and references therein. In brief, the wave-front model is run for a set of training points; the emulator then allows the interpolation of

the model output between these points. For pragmatic reasons, we build an individual emulator for each radiocarbon site, rather than attempt to build a complex time-space emulator.

Consider the arrival time $\tau(\mathbf{x}_i|\boldsymbol{\theta})$ at a single site i . For simplicity of notation, we denote this arrival time by $\tau(\boldsymbol{\theta})$. Our emulator for the arrival time uses a Gaussian process with mean $m(\cdot)$ and covariance function $k(\cdot, \cdot)$, that is,

$$\tau(\cdot) \sim \text{GP}(m(\cdot), k(\cdot, \cdot)). \quad (\text{B1})$$

We choose a suitable form for the mean function, given the approximate relationship expected between the arrival time at a site and the parameter values, which are all speeds. The simple relation $d = Ut$, where d represents distance, t time, and U speed, gives $t \propto 1/U$, so we choose a mean function which reflects this:

$$m(\boldsymbol{\theta}) = \alpha_0 + \alpha_1 \frac{1}{U_0} + \alpha_2 \frac{1}{V_R} + \alpha_3 \frac{1}{V_C}, \quad (\text{B2})$$

where the coefficients α_k are determined using least-squares fits. These coefficients essentially account for the relative importance of diffusive, river, and coastal spread, given the complicated geography between the source of the spread and the particular site being emulated. There are various possible choices for the form of covariance function. We use a stationary Gaussian covariance function

$$k(\boldsymbol{\theta}, \boldsymbol{\theta}') = a \exp \left(- \sum_{j=1}^3 \frac{(\theta_j - \theta'_j)^2}{r_j^2} \right), \quad (\text{B3})$$

with hyperparameters a and r_j ($j = 1, 2, 3$), which must be determined from the training data.

Suppose that p simulations of the (computationally expensive) wave-front model are available to us, each providing the arrival time at each radiocarbon-dated site. Let $\boldsymbol{\tau}(\boldsymbol{\Theta}) = (\tau(\boldsymbol{\theta}_1), \dots, \tau(\boldsymbol{\theta}_p))^T$ denote the p vector of arrival times resulting from the wave-front model with input values $\boldsymbol{\Theta} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_p)^T$, where $\boldsymbol{\theta}_i = (U_{0,i}, V_{R,i}, V_{C,i})^T$. A Gaussian

process can be viewed as an infinite collection of random variables, any finite number of which are jointly normally distributed. Therefore, from Eq. (B1), we have

$$\boldsymbol{\tau}(\boldsymbol{\Theta}) \sim N(\mathbf{m}(\boldsymbol{\Theta}), K(\boldsymbol{\Theta}, \boldsymbol{\Theta})),$$

where $\mathbf{m}(\boldsymbol{\Theta})$ is the mean vector with j th element $m(\boldsymbol{\theta}_j)$, and $K(\boldsymbol{\Theta}, \boldsymbol{\Theta})$ is the variance matrix with (j, ℓ) th element $k(\boldsymbol{\theta}_j, \boldsymbol{\theta}_\ell)$.

We can model the front arrival time at the site for other values of the input parameters $\boldsymbol{\theta}^*$ as follows. Using the standard properties of the multivariate normal distribution, the arrival time has distribution

$$\tau(\boldsymbol{\theta}^*)|\boldsymbol{\tau}(\boldsymbol{\Theta}) \sim N(\mu(\boldsymbol{\theta}^*), \Sigma(\boldsymbol{\theta}^*)), \quad (\text{B4})$$

where

$$\begin{aligned} \mu(\boldsymbol{\theta}^*) &= m(\boldsymbol{\theta}^*) + K(\boldsymbol{\theta}^*, \boldsymbol{\Theta})[K(\boldsymbol{\Theta}, \boldsymbol{\Theta})]^{-1}[\boldsymbol{\tau}(\boldsymbol{\Theta}) - \mathbf{m}(\boldsymbol{\Theta})], \\ \Sigma(\boldsymbol{\theta}^*) &= K(\boldsymbol{\theta}^*, \boldsymbol{\theta}^*) - K(\boldsymbol{\theta}^*, \boldsymbol{\Theta})[K(\boldsymbol{\Theta}, \boldsymbol{\Theta})]^{-1}K(\boldsymbol{\Theta}, \boldsymbol{\theta}^*). \end{aligned}$$

To simplify the notation, we have dropped the dependence in these expressions on the hyperparameters a and r_j ($j = 1, 2, 3$).

1. Fitting the emulator

We build a separate emulator for each site for which we have radiocarbon data. Although a single run of the wave-front model for particular input parameters $\boldsymbol{\theta}_i$ is computationally intensive, such a run gives the first-arrival time at all sites, so that only p runs of the wave-front model are needed to construct the training data for all n emulators.

We fit each emulator using a Metropolis-Hasting algorithm (similar to that described in Appendix A), to obtain the posterior distributions for the hyperparameters. Figure 9 (left) shows the traces of the resulting hyperparameter chains (for a single radiocarbon site), and Fig. 9 (right) shows the corresponding posterior densities. These plots are representative of the MCMC estimation of the posterior hyperparameter distributions at other radiocarbon sites.

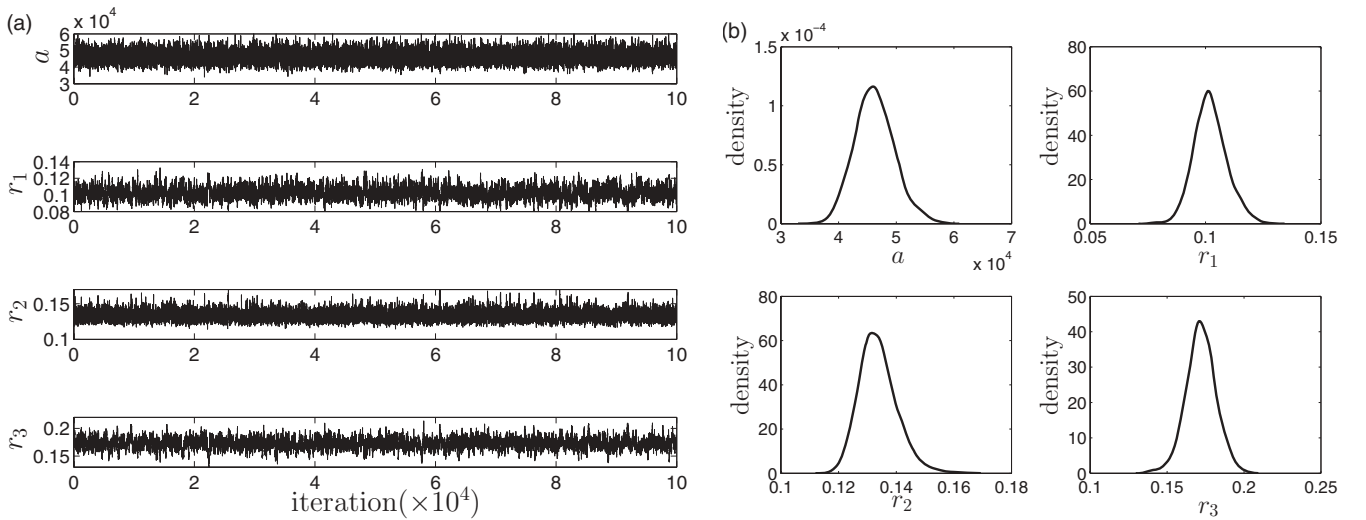


FIG. 9. Traces for the hyperparameters of the emulator for the radiocarbon site Achilleion, located at latitude 39.2° N, longitude 22.38° E (a), and the posterior distributions of the four hyperparameters from these MCMC chains (b). The output from these chains varies from site to site, and it is important to construct a separate emulator for each site, to obtain accurate emulation of the wave-front model.

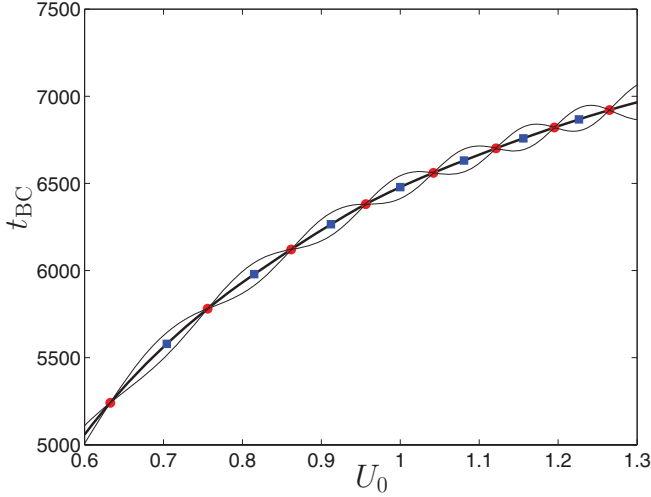


FIG. 10. (Color online) The mean output from one emulator (thick black line), with the training points used also plotted (red circles). The thin upper and lower lines show $\pm$ two standard deviations of the emulator output. The (blue) squares represent further output from the wave-front model, which were not used in constructing the emulator. The discrepancy between these points and the emulator output can be used as a test of the emulator.

While it is possible in theory to fit the emulator and the statistical model in Eq. (9) jointly using an MCMC scheme, this would be extremely computationally expensive. We therefore fit the emulator and the statistical model separately. In particular, when using the emulator output in the inference scheme described in Appendix A, we fix the hyperparameters at their posterior means. We believe this approach is justified, as even allowing for the (low) posterior uncertainty of the hyperparameters makes little difference to the (predictive) fit of the emulators.

For illustration, Fig. 10 shows the output from one emulator, together with the training data, when we fix two of the parameters (V_C and V_R) and consider only variations in the U_0 axis. The magnitudes of the errors shown in the plot are consistent with those from the three-parameter emulator.

2. Selection of training points

The selection of the training points Θ in parameter space merits further comment. Although using a regular lattice design is appealingly simple, it is not particularly efficient. Instead, we adopt a more commonly used design for fitting Gaussian processes, the latin hypercube design (LHD) [41]. Designs of this class distribute points within a hypercube in parameter space more efficiently than a lattice design. If we consider any single parameter direction in isolation, the mean separation between points is p^{-1} , as opposed to $p^{-1/3}$ for a regular lattice.

We constructed our 200-point LHD using the MATLAB routine `lhsdesign`. Initially we set the lower bounds of the hypercube to be the origin and used the upper first percentiles of the prior distribution as its upper bounds. We then repeatedly ran the inference algorithm (described in the following section), used the results to determine a conservative estimate of a hypercube containing all points in the MCMC

output (and therefore plausibly containing all of the posterior density), and generated another LHD. The final LHD used for inferences on $(U_0, V_C, V_R)^T$ in this paper is contained within the hypercube $(0, 3.1) \times (0, 3) \times (0, 2)$.

3. Testing the emulator

It is imperative that the accuracy of the fitted emulators as an approximation to the wave-front model be assessed. We therefore considered various quantitative statistics [42]. We created a second LHD with $p^* = 100$ points, $\Theta^* = (\theta_1^*, \dots, \theta_{p^*}^*)^T$, and determined the front arrival time at all sites for each θ_i^* using both the emulator mean (with its hyperparameters fixed at their posterior means) and the wave-front model: we denote these arrival times by τ^* and τ , respectively. (Separate values of these quantities exist for all sites; but for simplicity, as in the preceding sections, the specialization to individual sites is left implicit.)

For brevity, we discuss only the analysis of a statistic which includes both site-specific accuracy and correlation between residual errors (at the emulator test points): the Mahalanobis distance, D , defined via

$$D^2 = (\tau^* - \tau)^T V(\Theta^*)^{-1} (\tau^* - \tau), \quad (B5)$$

where

$$V(\Theta^*) = K(\Theta^*, \Theta^*) - K(\Theta^*, \Theta) K(\Theta, \Theta)^{-1} K(\Theta, \Theta^*).$$

[Note that $V(\Theta^*)$ is defined analogously to $\Sigma(\theta^*)$ above, but now contains information about all p^* test points in Θ^* .] It can be shown that D^2 follows a scaled F distribution [42] in the case of the GP emulator, with $D^2 \sim p^*(p-5)F_{p^*, p-3}/(p-3)$. Figure 11 shows the Mahalanobis distance at each site, together with the upper 95% point of its distribution. This, along with our analyses of other statistics (not presented here), confirms that the emulators provide a reasonable fit throughout the design space. These diagnostics gave similar results for different LHDs, without any systematic site-specific biases.

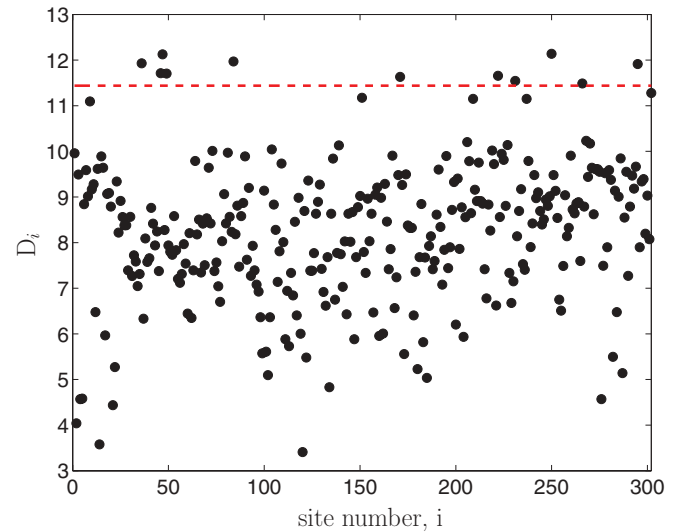


FIG. 11. (Color online) Test of the fit of the emulators to the radiocarbon data: the Mahalanobis distance D_i is plotted for each radiocarbon site i . The horizontal (red) dashed line (at $D_i = 11.44$) marks the upper 95% point of the D distribution.

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